

Lesson 3-5, Day 3 — Multiplicity & Factored-Form Graphing

Name: _____ Period: _____ Date: 5/8/26

Math Objective

Construct a factored polynomial from a graph by identifying x-intercepts, choosing the correct multiplicity at each (touch = even, cross = odd), and applying end behavior to determine the leading coefficient sign.

Language Objective

Students will use the frame “The factor for $x =$ _____ must be _____ because the graph _____” to justify each factor of a polynomial, and explain why even multiplicity produces a touch while odd multiplicity produces a cross.

Essential Understanding

A polynomial’s graph and its factored form encode the same information — zeros locate the factors, multiplicity determines the local behavior at each x-intercept (touch vs. cross), and end behavior determines the leading coefficient sign.

Sentence frames (use these when you share):

- “The factor for $x =$ _____ must be _____ because the graph _____.”
- “The leading coefficient is _____ because the ends go _____.”
- “I notice _____, so I think _____.”

Do Now (5 min)

Quick recall from yesterday.

1. What is the multiplicity of the zero $x = 4$ in $f(x) = (x - 4)^2(x + 1)$? _____

2. If a zero has **even** multiplicity, what does the graph do at that zero?

3. If a zero has **odd** multiplicity, what does the graph do at that zero?

Launch (15 min) — Gradual release

I do (4 min): watch and annotate

Teacher example: $f(x) = (x - 2)^2(x + 3)$.

Annotate the example as the teacher works through it. Write what each piece tells you:

Zero $x = 2$ multiplicity = _____, behavior at zero:

Zero $x = -3$ multiplicity = _____, behavior at zero:

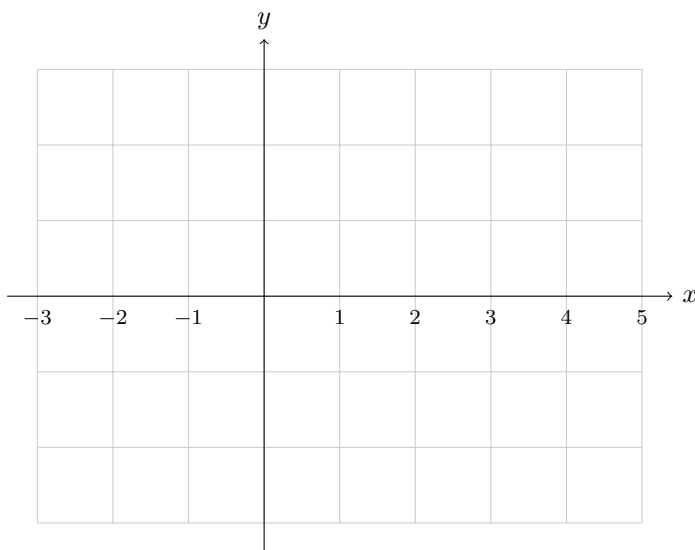
Degree _____, leading coefficient sign: _____

End behavior _____, _____
(left/right)

Touch / Cross rule (write what the teacher says):

You do (4 min): try this one yourself

Sketch $g(x) = (x + 1)(x - 4)^2$ on the grid below. Label both zeros and indicate which is a touch and which is a cross.



Sketch the shape only — exact y-values not required. Mark zeros and label touch/cross.

Why? _____

I check (3 min)

Compare your sketch with the projected exemplar. **What did you fix?**

We do together (4 min)

Class example: $h(x) = -x(x + 2)^2(x - 1)$. Fill in as we go.

• Zeros: $x =$ _____, _____, _____

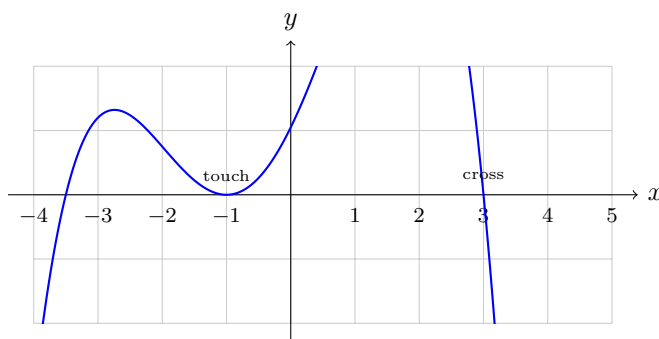
• Touch or cross at each? _____

- End behavior: left _____, right _____
(Why? _____)
-

Explore (25 min)

Mystery Graph (10 min, think-pair-share) — DOK 3

The graph below has zeros at $x = -1$ (touch) and $x = 3$ (cross). **Both ends point downward.**



(Note: this graph has more than two zeros — you only need to account for the two labeled. Focus on what those two tell you about factors and what the ends-down behavior tells you about leading coefficient and degree.)

Build a possible factored form for this graph. Justify every piece.

My factored form: $f(x) =$ _____

Why this factor for $x = -1$?

Why this factor for $x = 3$?

Why this leading coefficient?

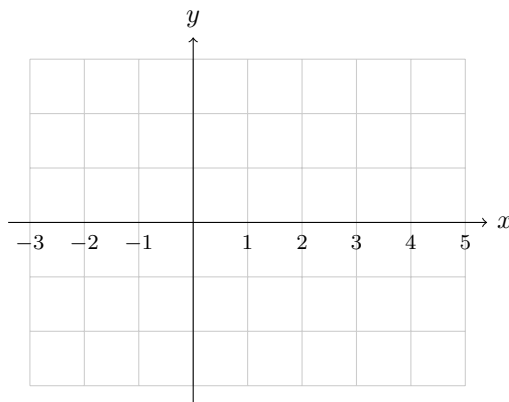
Could the degree be different and still match?

DOK-2 consolidation items (15 min, partners)

1. (Savvas q12) Sketch the graph of $f(x) = 3x^3 - 9x^2 - 12x$ by first finding its zeros.

Factored form: $f(x) =$ _____

Zeros: _____



Sketch the shape only — exact y-values not required. Mark zeros and label touch/cross.

2. (Try-It 2b) Describe the behavior of the graph of $f(x) = (x^2 + 9)(x - 1)^5(x + 2)^2$ at each of its zeros.

- Real zeros: _____
- At $x = 1$: multiplicity _____, touch / cross? _____
- At $x = -2$: multiplicity _____, touch / cross? _____
- What about the factor $(x^2 + 9)$?

3. A polynomial graph crosses at $x = 0$, touches at $x = 2$, and both ends point upward.

Write a possible factored form: $f(x) =$ _____

Justify the leading coefficient sign:

Share / Summary (5 min)

Today's rule, in your words:

One way the rule helped me on items 1–3 today:

Exit Ticket (5 min)

1. A factor of $(x - 4)^2$ produces a _____ at $x = 4$. (touch / cross)

2. A factor of $(x + 1)$ produces a _____ at $x = -1$. (touch / cross)

3. The biggest thing I learned today was:
